this convention whenever a result requires a single coordinate; the exchange principle and Lemma 2.15 below are false for arbitrary tuples.

From here on we work in a sufficiently saturated model $\mathbb{U}$ of $\mathrm{ACVF}(0,0)$, together with a parameter set $\mathbb{S}$, which is supposed to be a substructure of $\mathbb{U}$. So by a definable set we shall mean an $\mathbb{S}$-definable set, unless indicated otherwise.

Notation 2.2. There is a special element $\infty = \mathrm{rv}(0)$ in the RV-sort. For simplicity, we shall write RV to mean the RV-sort without the element ∞, and RV_∞ otherwise — that is, $\mathrm{RV} = \mathrm{rv}(\mathrm{VF}^\times)$ and $\mathrm{RV}_\infty = \mathrm{rv}(\mathrm{VF})$ — although the difference rarely matters (when it does we will of course provide further clarification).

Terminology 2.3 (Sets and subsets). By a definable set in VF we mean a definable subset in the sort VF, by which we just mean a subset of VF^n for some n, unless indicated otherwise; similarly for other sorts or structures in place of VF that have been clearly understood in the context, such as RV_∞, $\Bbbk$, $\mathcal{M}$, or any substructure $\mathbb{M}$ of $\mathbb{U}$. In particular, a definable set without further qualification means a definable set in $\mathbb{U}$, that is, a definable subset of $\mathrm{VF}^n \times \mathrm{RV}_\infty^m$ for some $n, m \in \mathbb{N}$.

Terminology 2.4 (Relations as functions). We may think of a relation C between two sets A and B, that is, a subset $C \subseteq A \times B$, as a function from A into the powerset $\mathcal{P}(B)$. For instance, if $A \subseteq \mathrm{VF}^n \times \mathrm{RV}^m$ is a definable set then it may be thought of as a *definable* function from A_{VF} into $\mathcal{P}(\mathrm{RV}^m)$. Thus a definable function $f : A \longrightarrow \mathcal{P}(B)$ is ontologically the same as the definable set $\bigcup_{a\in A} a \times f(a)$. We say that f is *finitary* if $f(a)$ is finite for every $a \in A$.

The default topology in $\mathbb{U}$ is the valuation topology.

Notation 2.5. A set $\mathfrak{p} \subseteq \mathrm{VF}^n \times \mathrm{RV}_\infty^m$ of the form $\prod_{i\leq n} \mathfrak{b}_i \times t$ is called an (*open, closed,* RV-) *polydisc* if each $\mathfrak{b}_i$ is an (open, closed, RV-) disc. The *radius* $\mathrm{rad}(\mathfrak{p})$ of the polydisc $\mathfrak{p}$ is the tuple $(\mathrm{rad}(\mathfrak{b}_1), \ldots, \mathrm{rad}(\mathfrak{b}_n))$. The open polydisc centered at $(a,t) \in \mathrm{VF}^n \times \mathrm{RV}_\infty^m$ with radius $\gamma \in \Gamma^n$ is denoted by $\mathfrak{o}((a,t),\gamma)$; the closed polydisc with radius $\gamma \in \Gamma_\infty^n$ is denoted by $\mathfrak{c}((a,t),\gamma)$. Radius ∞ is used only for closed discs, with $\mathfrak{c}(a,\infty) = \{a\}$; open discs have finite valuative radius.

The subdiscs $\mathfrak{o}(0,\gamma)$, $\mathfrak{c}(0,\gamma)$ of VF are more suggestively denoted by $\mathcal{M}_\gamma$, $\mathcal{O}_\gamma$, respectively, and $\mathcal{M}_0$, $\mathcal{O}_0$ are even more suggestively denoted by $\mathcal{M}$, $\mathcal{O}$. Write $\mathrm{RV}_\infty^{\circ\circ} = \mathrm{rv}(\mathcal{M})$ and $\mathrm{RV}^{\circ\circ} = \mathrm{RV}_\infty^{\circ\circ} \smallsetminus \infty$. More generally, for $\gamma \in \Gamma$, denote by $\mathrm{RV}_\gamma^{\circ\circ}$ the set $\mathrm{rv}(\mathcal{M}_\gamma \smallsetminus 0)$.

An RV-*disc* is a fiber $\mathrm{rv}^{-1}(t)$ for $t \in \mathrm{RV}_\infty$, including $\mathrm{rv}^{-1}(\infty) = \{0\}$. In a mixed ambient space, an RV-polydisc keeps the existing RV_∞-coordinates fixed. The RV-*hull* of a set A, denoted by $\mathrm{RVH}(A)$, is the union of all the RV-polydiscs whose intersections with A are nonempty. If A equals $\mathrm{RVH}(A)$ then A is called an RV-*pullback*.

Resolved referee R022 [Clarification] – RV-discs and the zero fiber.
Original source line 748. Define an RV-disc explicitly as $\mathrm{rv}^{-1}(t)$ for $t \in \mathrm{RV}_\infty$, including $\mathrm{rv}^{-1}(\infty) = \{0\}$. This convention is needed for the equivalence between being an RV-pullback and having empty RV-boundary; otherwise sets meeting coordinate hyperplanes are not covered by their stated RV-hulls. Also specify that taking the hull keeps the existing RV-coordinates fixed.

Author resolution. This is the intended convention. The definition now explicitly includes the singleton zero fiber and preserves the existing RV-coordinates under the hull.

Terminology 2.6. A VF-*fiber* of a set A is a set of the form A_t, where $t \in A_{\mathrm{RV}}$; in particular, a VF-fiber of a function $f : A \longrightarrow B$ is a set of the form f_t for some $t \in f_{\mathrm{RV}}$ (here f also stands for its own graph) which is indeed (the graph of) a function. We say that A is open if every one of its VF-fibers is, f is continuous if every one of its VF-fibers is, and so on. Composition is understood fiberwise,

retaining the intermediate RV-coordinates: for every fixed input, intermediate, and output RV-tuple, the corresponding VF-fiber maps compose continuously on their natural domain. Thus, if f, g are continuous functions with $\mathrm{ran}(f) \subseteq \mathrm{dom}(g)$, the composite with these intermediate labels retained is continuous in this fiberwise sense.

Resolved referee R023 [Error] – Fiberwise continuity is not closed under composition. *Original source line 753.* The last assertion is false with the stated definition. Put $r = \mathrm{rv}(1)$ and $s = \mathrm{rv}(-1)$, let $f : \mathrm{VF} \longrightarrow \{r, s\}$ send 0 to r and every nonzero point to s, and let $g(r) = 0$, $g(s) = 1$ in VF. Every VF-fiber of both graphs is continuous in the prescribed sense, whereas $g \circ f$ is discontinuous at 0. A sufficient repair is to require, in addition, that the output RV-coordinates be locally constant on each input VF-fiber; equivalently use the valuation topology on VF and the discrete topology on RV. If the weaker convention is retained, delete the composition assertion and check every later use of composition separately.

Author resolution. The composition remark concerns composition on fixed RV-fibers, with the intermediate RV-labels retained. This intention is now explicit. The original fiberwise continuity convention is retained; no stronger topology is imposed. The example discards the intermediate label, so it does not contradict the clarified assertion.

Notation 2.7. Semantically we shall treat the value group Γ as a definable sort (the Γ-sort). However, syntactically any reference to Γ may be eliminated in the usual way and we can still work with $\mathcal{L}_{\mathrm{RV}}$-formulas for the same purpose. This implies that $\gamma \in \Gamma$ is definable means that $\mathrm{val}^{-1}(\gamma)$ is definable.

We shall write $\gamma^\sharp$, $\gamma \in \Gamma$, when we want to emphasize that it is the set $\mathrm{vrv}^{-1}(\gamma) \subseteq \mathrm{RV}$ that is being considered. More generally, if I is a set in Γ then we write $I^\sharp = \bigcup\{\gamma^\sharp \mid \gamma \in I\}$. Similarly, if U is a set in RV then $U^\sharp$ stands for $\bigcup\{\mathrm{rv}^{-1}(t) \mid t \in U\}$. In particular, $\gamma^{\sharp\sharp} = \{x \in \mathrm{VF} \mid \mathrm{val}(x) = \gamma\}$.

Notation 2.8. Let $U \subseteq \mathrm{RV}^n \times \Gamma^m$, $V \subseteq \mathrm{RV}^{n'} \times \Gamma^{m'}$, and $C \subseteq U \times V$. The *Γ-Jacobian* of C at $((u,\alpha),(v,\beta)) \in C$, written as $\mathrm{Jcb}_\Gamma((u,\alpha),(v,\beta))$, is the element

$$\mathrm{Jcb}_\Gamma((u,\alpha),(v,\beta)) = -\Sigma(\mathrm{vrv}(u),\alpha) + \Sigma(\mathrm{vrv}(v),\beta),$$

where $\Sigma(\gamma_1,\ldots,\gamma_n) = \gamma_1 + \ldots + \gamma_n$. If C is the graph of a function then we just write $\mathrm{Jcb}_\Gamma C(u,\alpha)$ instead of $\mathrm{Jcb}_\Gamma((u,\alpha), C(u,\alpha))$.

Convention 2.9. We may and shall assume that in any $\mathcal{L}_{\mathrm{RV}}$-formula, every VF-sort term (polynomial) occurs in the scope of an instance of the function symbol rv. For example, if $f(x)$, $g(x)$ are polynomials then the formula $f(x) = g(x)$ is equivalent to $\mathrm{rv}(f(x) - g(x)) = \infty$. The polynomial $f(x)$ in $\mathrm{rv}(f(x))$ is referred to as a *top term* (of the $\mathcal{L}_{\mathrm{RV}}$-formula in question).

Notation 2.10 (The definable sort DC of discs). At times it will be more convenient to work in the traditional expansion $\mathbb{U}^{\mathrm{eq}}$ of $\mathbb{U}$ by all definable sorts. However, for our purpose, a much simpler expansion $\mathbb{U}^\bullet$ suffices. This expansion has only one additional sort DC that contains, as elements, all the open and closed discs. Since each point in VF may be regarded as a closed disc of valuative radius ∞, for convenience, we may and occasionally do think of VF as a subset of DC. Heuristically, we may think of a disc that is properly contained in an RV-disc as a "thickened" point of certain stature in VF. There are two uniform cross-sort maps $\mathrm{VF} \times \Gamma \longrightarrow \mathrm{DC}$ in $\mathbb{U}^\bullet$, sending (a,γ) to the open disc $\mathfrak{o}(a,\gamma)$ and to the closed disc $\mathfrak{c}(a,\gamma)$, respectively. The closed-disc map extends to $\mathrm{VF} \times \Gamma_\infty$ by $(a,\infty) \mapsto \{a\}$. The radius is an argument of these maps, rather than an index for separately named function symbols.

The expansion $\mathbb{U}^\bullet$ can help reduce the technical complexity of our discussion. However, as is the case with the definable Γ-sort, it is conceptually inessential since, for the purpose of this paper, all

allusions to discs as (imaginary) elements may be eliminated in favor of objects already definable in $\mathbb{U}$.

Whether parameters in DC are used or not shall be indicated explicitly, if it is necessary, by means of introducing parameters of the form $\ulcorner \mathfrak{a} \urcorner$ or the phrase "in $\mathbb{U}^\bullet$," in which case the substructure $\mathbb{S}$ may contain names for discs that may or may not be definable from $\mathrm{VF}(\mathbb{S}) \cup \mathrm{RV}(\mathbb{S})$.

A disc centered at 0 may be identified with its valuative radius together with its open/closed designation (Γ is already treated as a definable sort). At a fixed finite radius the open and closed discs are distinct; the radius ∞ designates only the closed singleton disc.

For a disc $\mathfrak{a} \subseteq \mathrm{VF}$, the corresponding imaginary element in DC is denoted by $\ulcorner \mathfrak{a} \urcorner$ when notational distinction makes the discussion more streamlined; $\ulcorner \mathfrak{a} \urcorner$ may be heuristically thought of as the "name" of $\mathfrak{a}$. Conversely, a set $D \subseteq \mathrm{DC}$ is often identified with the set $\{\mathfrak{a} \mid \ulcorner \mathfrak{a} \urcorner \in D\}$, in which case $\bigcup D$ denotes a subset of VF.

Resolved referee R024 [Clarification] – A uniform language for the disc sort.
Original source line 791. For a literal language expansion, use two uniform maps $(a, \gamma) \mapsto \mathfrak{o}(a, \gamma)$ and $(a, \gamma) \mapsto \mathfrak{c}(a, \gamma)$, with domain $\mathrm{VF} \times \Gamma$ (and the specified degenerate cases). A separate named function for every element γ of the monster would name those radii and change the parameter structure. Also retain an open/closed tag when identifying discs centered at 0 with radii: the two discs with a fixed finite radius are distinct.

Author resolution. The original wording was informal shorthand for uniform maps with the radius as an argument. The maps and the open/closed designation are now explicit. Open discs have finite valuative radius; only closed discs admit the singleton radius-∞ case.

2.2. **Of points, discs, and atoms.** The substructure $\mathbb{S}$ is VF-*generated* if it is generated solely by elements in VF, similarly for being $(\mathrm{VF}, \mathrm{RV})$-generated, and so on. If $\mathbb{S}$ is VF-generated then the map rv is surjective in $\mathbb{S}$, but its value group $\Gamma(\mathbb{S})$ could still be trivial. If $\Gamma(\mathbb{S})$ is indeed nontrivial then the model-theoretic algebraic closure $\operatorname{acl}\mathbb{S}$ of $\mathbb{S}$ is a model of $\mathrm{ACVF}(0,0)$; see, for instance, [16, Corollary 4.18]. In that case, we also know what the definable closure $\operatorname{dcl}\mathbb{S}$ of $\mathbb{S}$ is:

Lemma 2.11. *Suppose that* $\mathbb{S}$ *is* VF*-generated and* $\Gamma(\mathbb{S})$ *is nontrivial. Then* $\operatorname{dcl}\mathbb{S} = \mathbb{S}$ *if and only if the underlying valued field* $(\mathrm{VF}(\mathbb{S}), \mathcal{O}(\mathbb{S}))$ *of* $\mathbb{S}$ *is henselian.*

Proof. We have $\mathrm{VF}(\operatorname{acl}\mathbb{S}) = \mathrm{VF}(\mathbb{S})^{\mathrm{ac}}$, where the latter is the field-theoretic algebraic closure of $\mathrm{VF}(\mathbb{S})$. Since the valued field automorphisms of $(\mathrm{VF}(\mathbb{S})^{\mathrm{ac}}, \mathcal{O}(\operatorname{acl}\mathbb{S}))$ over $(\mathrm{VF}(\mathbb{S}), \mathcal{O}(\mathbb{S}))$ are in one-to-one correspondence with the $\mathcal{L}_{\mathrm{RV}}$-automorphisms of $\operatorname{acl}\mathbb{S}$ over $\mathbb{S}$ and $\mathrm{VF}(\mathbb{S})$ is the fixed field with respect to these automorphisms if and only if $(\mathrm{VF}(\mathbb{S}), \mathcal{O}(\mathbb{S}))$ is henselian, we see that $\mathrm{VF}(\operatorname{dcl}\mathbb{S}) = \mathrm{VF}(\mathbb{S})$ if and only if $(\mathrm{VF}(\mathbb{S}), \mathcal{O}(\mathbb{S}))$ is henselian. By quantifier elimination and routine syntactical inspection, $\mathrm{RV}(\operatorname{dcl}\mathbb{S}) = \mathrm{RV}(\mathbb{S})$. The lemma follows. □

We shall impose at a later stage the assumption that $\mathbb{S}$ is VF-generated and (without loss of generality) definably closed, that is, $\operatorname{dcl}\mathbb{S} = \mathbb{S}$.

Lemma 2.12 ([16, Lemma 4.10])**.** *For any finite definable set* A *in* VF *there is a definable bijection* $f : A \longrightarrow U$ *for some set* U *in* RV*.*

The phenomenon exemplified by this simple lemma is dubbed "internalizing finite sets." Its failure in positive residue characteristics belies the deep obstacles one faces in developing a Hrushovski-Kazhdan integration theory therein; see the discussion around [8, Lemma 3.9].

Lemma 2.13 ([17, Lemma 3.3])**.** *For* $a, b \in \mathrm{VF}$ *and* $t \in \mathrm{RV}$*, if* b *is algebraic over* (a, t) *then it is algebraic over* a*.*